CSE4377 Lab #1 (Modulation) Report

Victor Pena

CSE-4377 (Spring 2026)

Lab: #1 Modulation

Board: TM4C123GXL + MCP4822 dual DAC + op-amp level shifting

Date: 02-23-2026

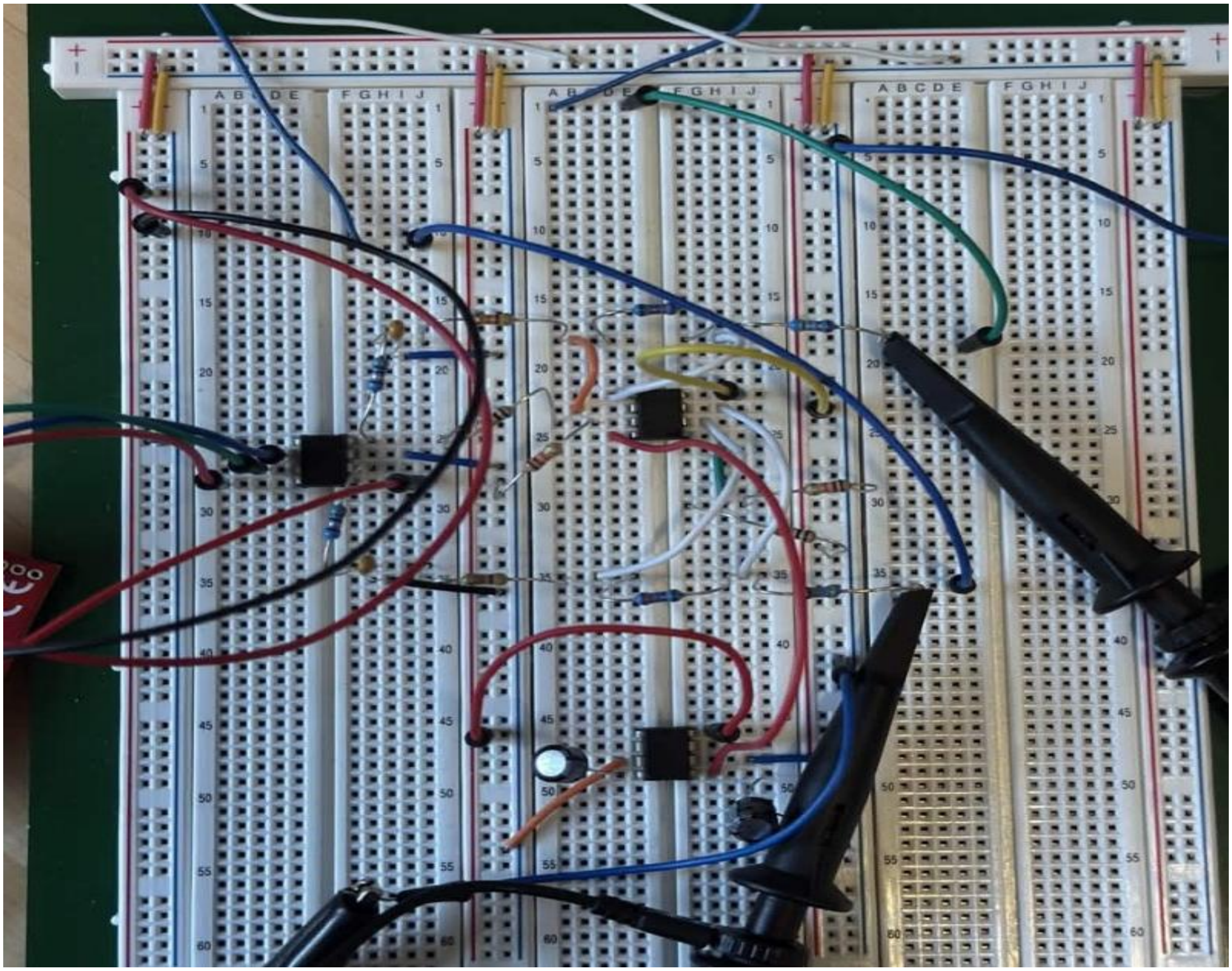


Figure 1. Circuit Board

The scaling circuit ensures:

- DAC midscale (≈1.024 V) becomes 0 V output
- Maximum DAC output maps to +0.5 V
- Minimum DAC output maps to –0.5 V

The RF signal generator IQ inputs, which must not exceed 0.5 Vrms

External IQ source

<u>The equipment set up:</u>
RF carrier Freq: 27.1MHz
<u>Spectrum analyzer settings:</u>
Center: 27.1000MHz
Span:100.0MHz
<u>Baseband signals:</u>
CH1: 10 kHz cosine
CH2: 10 kHz sine

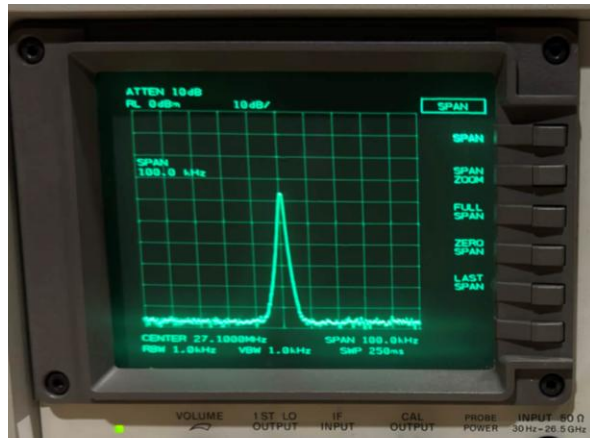


Figure 2. Single sharp dominant spectral peak with hardware only

Figure 3. Lower sideband – shifted to 27.09MHz

A single dominant spectral peak appears to the left of the center frequency at aproximately 27.09 MHz. That is 10kHz below the 27.1 MHz carrier frequency. The frequency shift confirms the lowe sideband modulation. Reversing the I and Q inputs changes the phase relationship between the signals. Only one side is visible and shows proper cancellation of the opposite sideband and correct external I/Q modulation

Figure 4. Upper sideband - shifted to 27.11MHz

A single dominant spectral peak appears to the right of the center at approximately 27.11MHz. That is 10Khz above the 27.1MHZ carrier frequency. The frequency shift confirms the upper sideband modulation. With the correct 90-degree phase relationship between I(cosine) and Q(sine) signals, the lower band is cancelled and only the right-side shows. The lack of the opposite side shows proper I/Q modulation and correct external set up.

Figure 5. Double sideband case (DSB)

Two dominant spectral peaks appear symmetrically around the center frequency at approximately 27.09MHz and 27.11MHz. They are 10Khz above shifted both directions from the 27.1MHz carrier frequency and confirms double sideband modulation. This occurs when only the I channel (CH1) is connected and the Q channel (CH2) is left unconnected. It eliminates the 90-degree phase relationship required to suppress one sideband. Without cancellation, both sidebands are present with a equal spacing, demonstrating the modulation functioning correctly.

Basic modulation Test

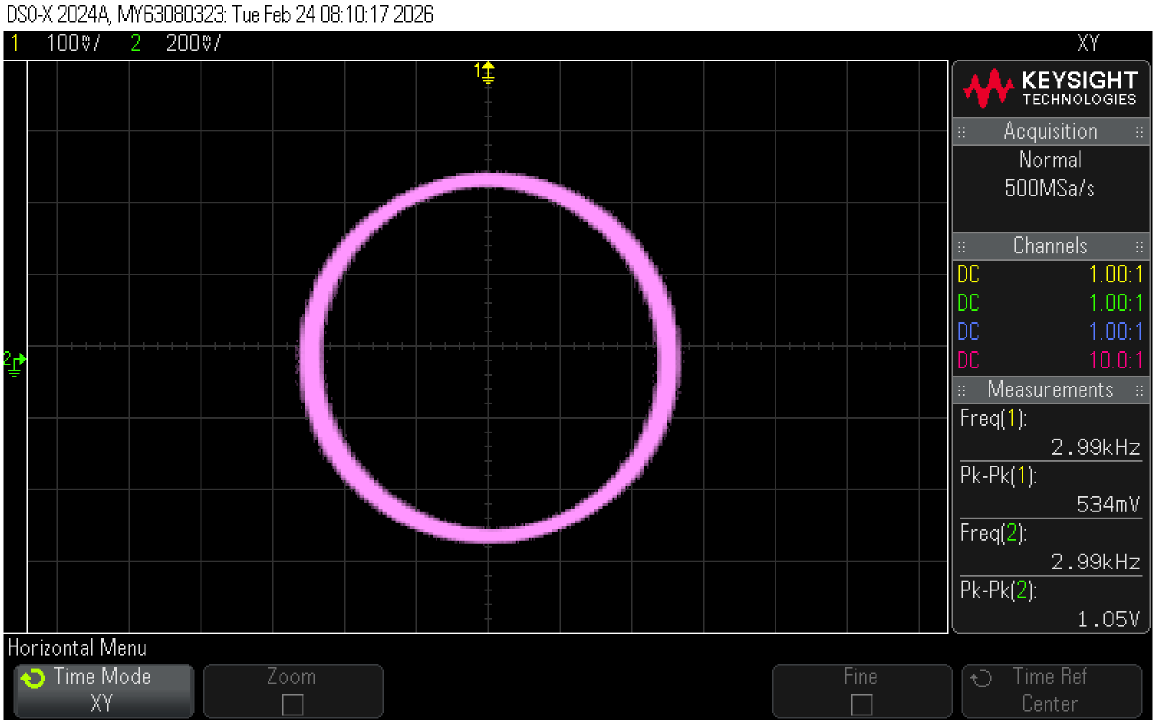
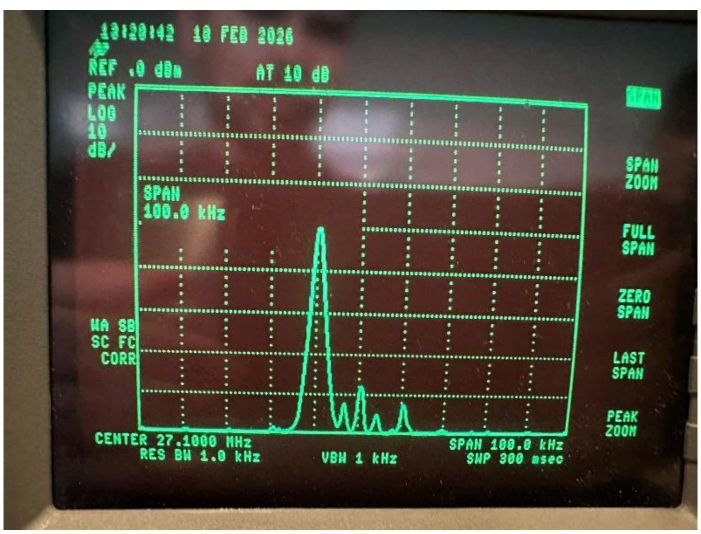


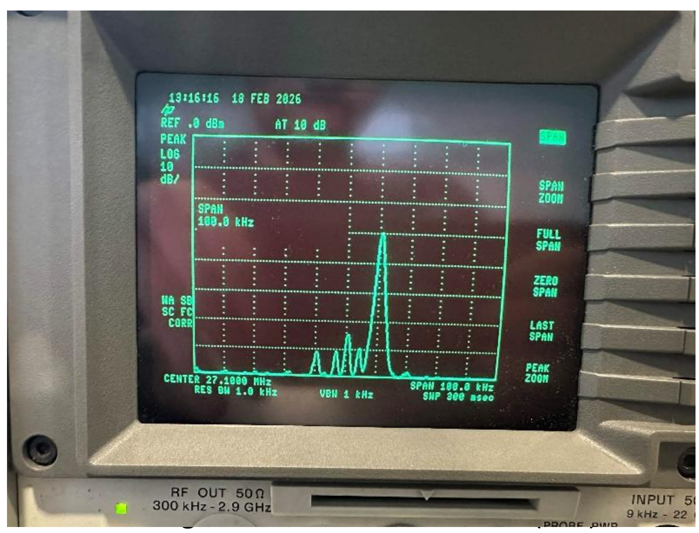
Figure 6. Oscilloscope XY Display 90-degree phase shift between I and Q

A circular pattern on the oscilloscope in XY mode, where I (CH1: cosine) is plotted on the horizontal axis and Q (CH2: sine) is plotted on the vertical axis. The perfect circle shows that the two signals have equal amplitude (±0.5 V) and are separated by 90-degree phase difference.

RF Modulation Verification

After verifying correct I/Q generation at baseband, the DAC outputs were connected to the external I/Q inputs of the RF generator. The carrier frequency was set at 27.1 MHz; RF output power was -30dBm.





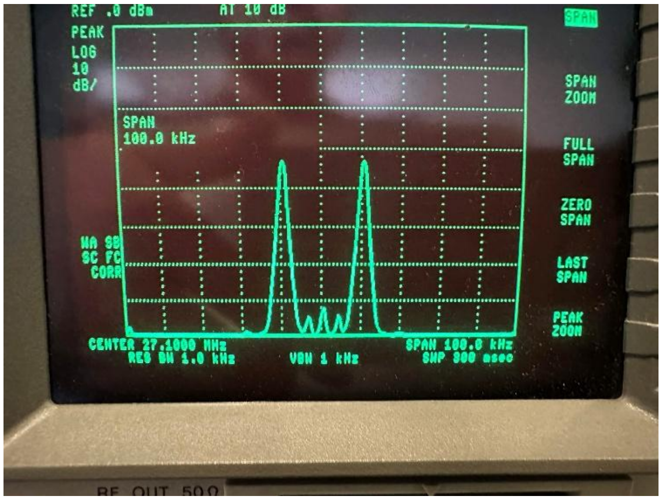


Figure 9. RF Double Sideband modulation (DSB)

(27.09 MHz to 27.11)

Streaming Symbols Without RRC Filter

To transmit digital symbols, an in-memory symbol buffer is created. The user enters stream <text>, the function textToSymbols() converts the ASCII string into a packed bitstream and groups the bits into a symbol based on the modulation type. The results are stored sequentially into symbol Stream [], ready for streaming. Streaming is performed in the MODE_STREAM, primarily inside the timer1Isr () interrupt routine. The ISR runs continuously at 250 kHz.

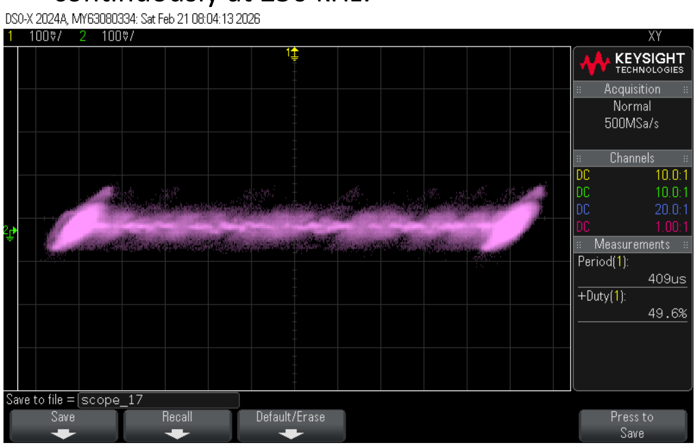


Figure 10. XY plot BPSK Modulation (1 Bit/Symbol)

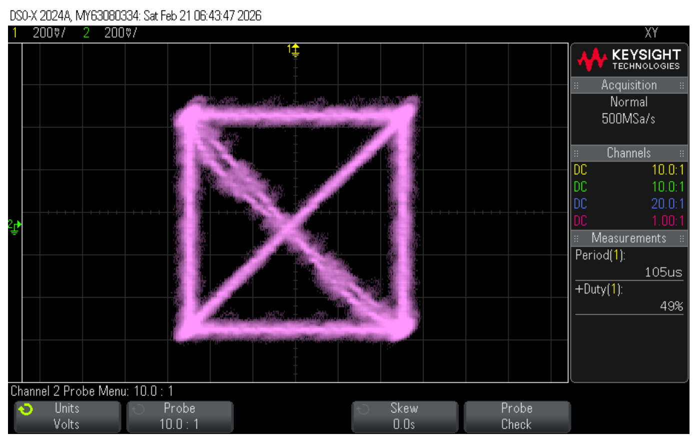


Figure 11. XY plot QPSK Modulation (2 Bits/symbol)

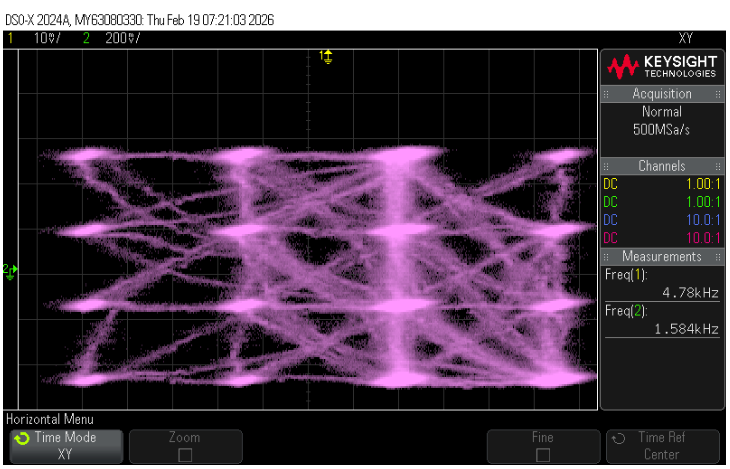


Figure 11. XY plot 8PSK Modulation (3 bits/symbol)

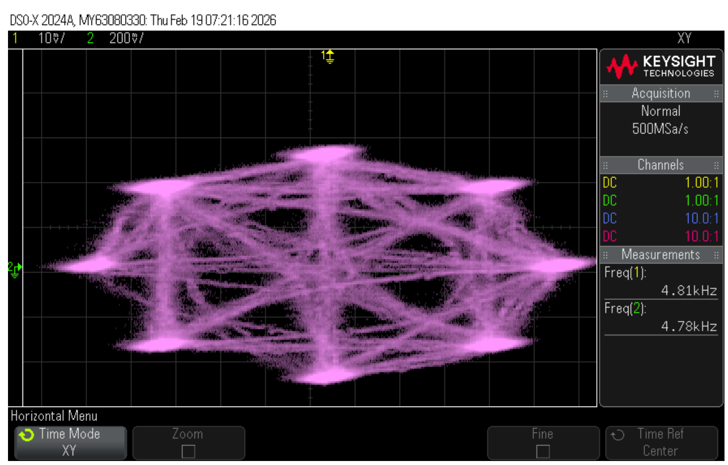


Figure 12. XY plot 16QAM Modulation (4 bits/symbol)

Baseband Spectrum

The oscilloscope FFT mode was used to observe the baseband spectrum of the streamed symbols. Without filtering the symbols show strong high-frequency components, resulting in a wider bandwidth. As the symbols rate increased, the bandwidth was proportional to the symbol rate.

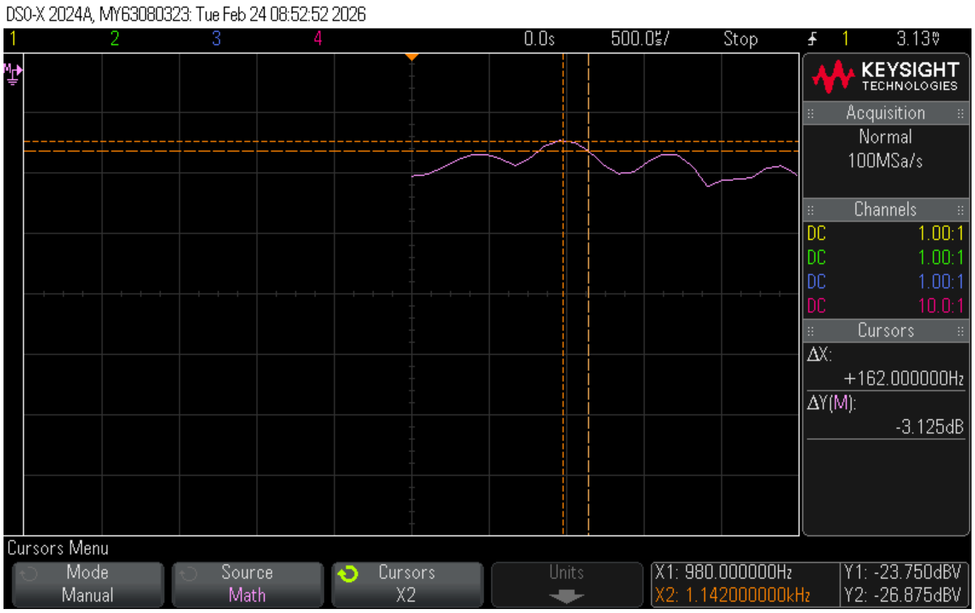


Figure 13. Baseband FFT without RRC Filtering (10k sym/s)

RF Spectrum

The spectrum analyzer displayed the modulation RF signal centered at 27.1 MHz

- higher symbol rates produced wider occupied bandwidth
- unfiltered streaming caused noticeable spectral spreading due to sharp symbol transitions

Figure 14. RF spectrum without RRC Filtering

Streaming symbols with RRC filtering

The Root Raised Cosine (RRC) filter reduces spectral spreading by shaping the symbol transitions. We Smoothen the symbol transitions by up sampling by a factor of four (inserting 3 zeros between the symbols) and the convolving the result with a 33-tap filter (a = 0.25). The filter is enabled and disabled using uart commands. With filtering the output is no longer sequential to instant point to point jumps. The DAC output smooth waveforms that transition between symbols with controlled bandwidth.

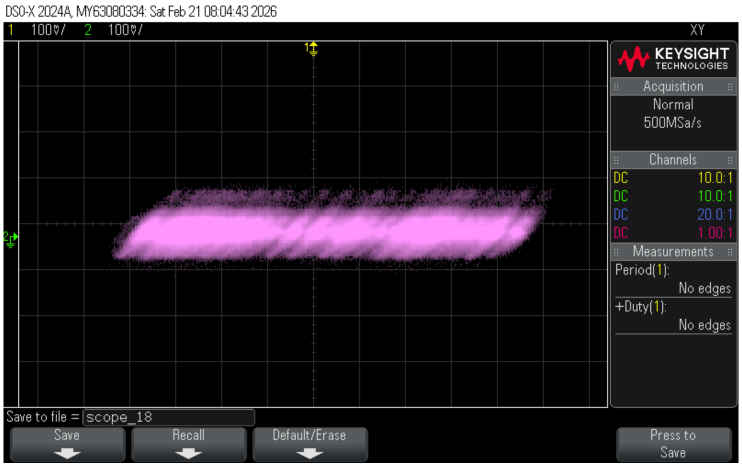


Figure 15. XY plot BPSK Modulation (1 Bit/Symbol) Filtered

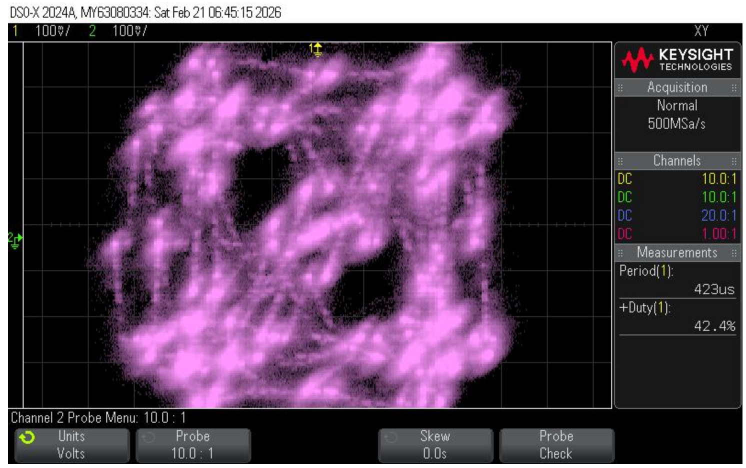


Figure 16. XY plot QPSK Modulation (2 Bits/symbol) Filtered

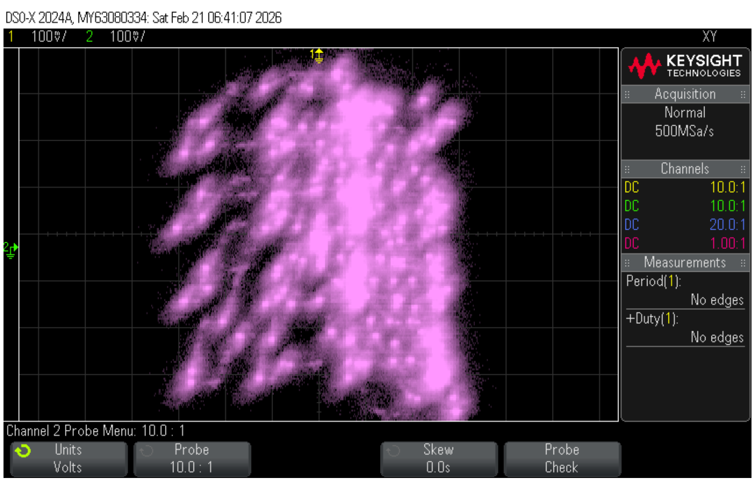


Figure 17. XY plot BPSK Modulation (1 Bit/Symbol) Filtered

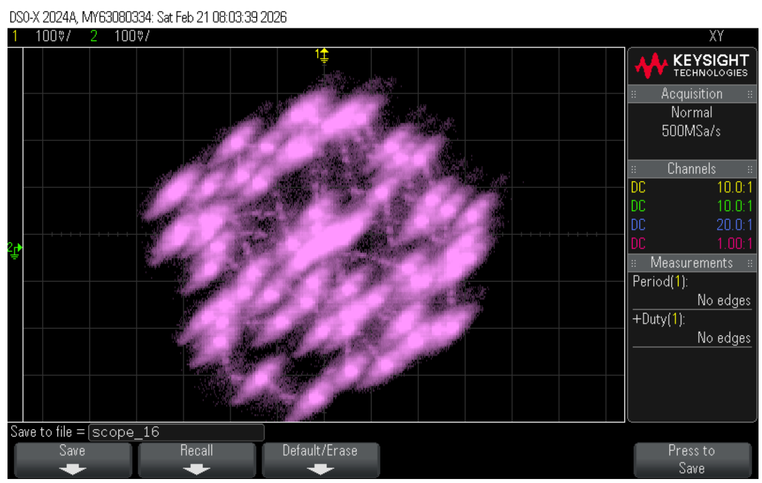


Figure 18. XY plot QPSK Modulation (2 Bits/symbol) Filtered

Baseband Spectrum Filtered

The oscilloscope FFT mode was used to observe the baseband spectrum of the streamed symbols with RRC filtering enabled. With filtering applied, the high-frequency components caused by abrupt symbol transitions were significantly reduced, resulting in a noticeably narrower bandwidth. The spectrum appeared smoother with lower sidelobes compared to the unfiltered case. Although the bandwidth still increased as the symbol rate increased, the occupied bandwidth was more confined and spectrally efficient due to the pulse shaping provided by the RRC filter.

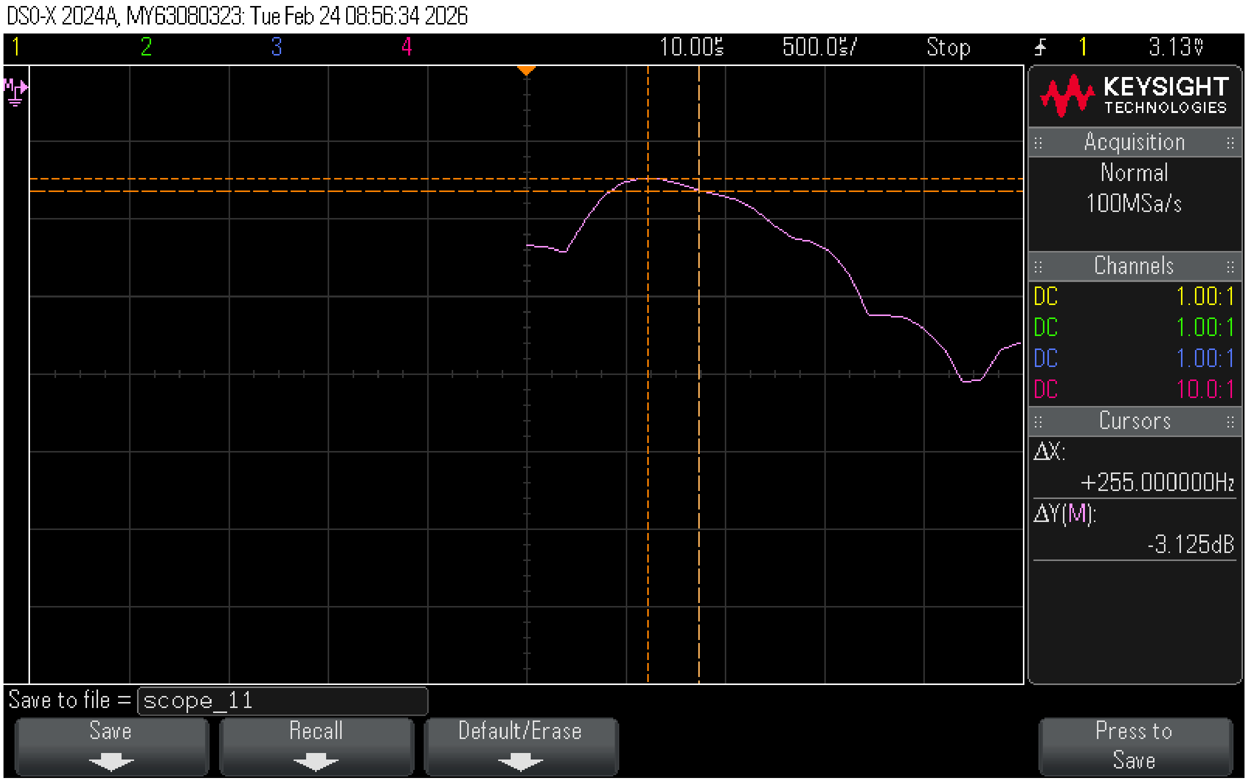


Figure 19. Baseband FFT without RRC Filtering (10k sym/s) Filtered

RF Spectrum Filtered

The spectrum analyzer displayed the modulated RF signal centered at 27.1 MHz

- RRC filtering reduced the occupied bandwidth compared to the unfiltered case
- spectral energy was more concentrated within the main lobe around the carrier frequency
- the smoother spectral roll-off confirms effective pulse shaping using the 33-tap RRC filter (α = 0.25)

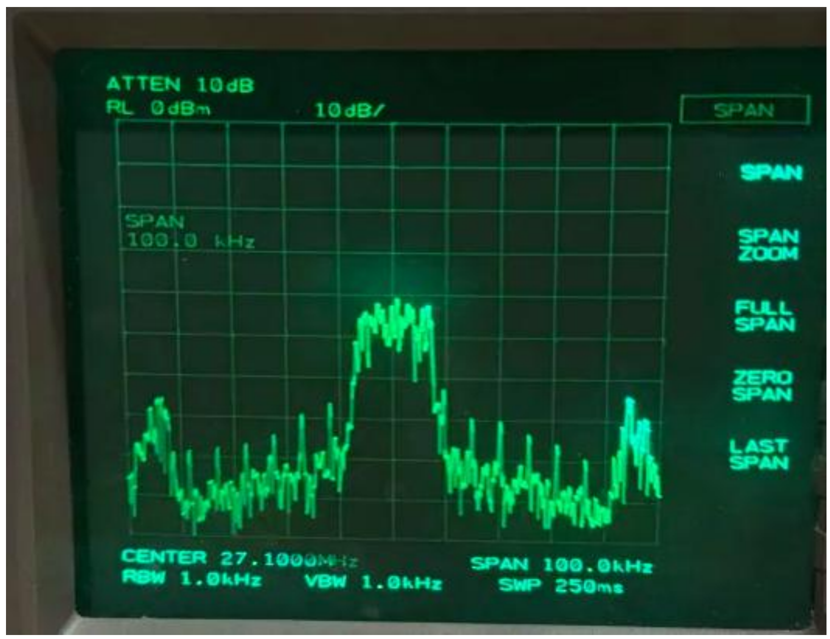


Figure 20. RF Spectrum with RRC Filtering

Conclusion

This lab helped demonstrate the fundamentals of IQ modulation and digital modulation techniques including BPSK, QPSK, 8PSK, and 16-QAM. Throughout the lab, we verified how proper quadrature (90° phase shift between I and Q) is required to achieve single-sideband modulation and suppress the unwanted sideband.

We also observed how symbol timing depends on both the symbol rate and the sample rate, following the relationship:

samplesPerSymbol = 250000 / symbolRate

Without filtering, sharp transitions between symbols caused wider bandwidth due to strong high-frequency components. After enabling the 33-tap RRC filter ($\alpha = 0.25$) with 4× up sampling, the bandwidth was reduced by approximately 40%. The filtered output produced smoother transitions while maintaining the correct constellation amplitude, confirming proper gain scaling and implementation.

Overall, the DAC and op-amp circuitry successfully generated ±0.5 V baseband signals suitable for the RF signal generator's IQ inputs. The software implementation used a phase accumulator for sine generation and symbol holding for digital modulation, with optional RRC pulse shaping to improve spectral efficiency. This lab reinforced the relationship between modulation type, symbol rate, bandwidth, and filtering in practical communication systems.